Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A bullet is fired normally on an immovable wooden plank. It loses 25% of its momentum in penetrating a thickness of 3.5 cm. The total thickness penetrated by the bullet is

(A) 8 cm

(B) 10 cm

(C) 12 cm

(D) 14 cm

Q.2

A bullet of mass m is fired horizontally with a velocity v on a wooden block of mass M suspended from a support and gets embedded in it. The kinetic energy of the bullet $+$ block system is

(A) $\frac{1}{2}mv^2$

(B) $\frac{1}{2}(M+m)v^2$

(C) $\frac{Mmv^2}{2(M+m)}$

(D) $\frac{m^2v^2}{2(M+m)}$

Q.3

A boy of mass $m = 40$ kg is standing on a stationary long plank of mass $M = 260$ kg floating on still water in a lake. He starts running with a velocity $v = 6$ ms^{-1} relative to the plank. Find the velocity of the boy relative to a stationary observer on the bank of the lake.

(A) $4.2\,\hat{\imath}$ms^{-1}

(B) $6.2\,\hat{\imath}$ms^{-1}

(C) $5.2\,\hat{\imath}$ms^{-1}

(D) $7.2\,\hat{\imath}$ms^{-1}

Q.4

A solid cylinder of mass M and radius R rolls down an inclined plane of height h. The angular velocity of the cylinder when it reaches the bottom of the plane will be

(A) $\frac{2}{R}\sqrt{gh}$

(B) $\frac{2}{R}\sqrt{\frac{gh}{2}}$

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(C) $\frac{2}{R}\sqrt{\frac{gh}{3}}$

(D) $\frac{1}{2R}\sqrt{gh}$

Q.5

A car of mass $1000\ \text{kg}$ is moving with a velocity of $10\ \text{ms}^{-1}$ towards another car of mass $1500\ \text{kg}$ moving with a velocity of $15\ \text{ms}^{-1}$ in the same direction. Find the velocity of the centre of mass of the two cars.

(A) $13\ \text{ms}^{-1}$

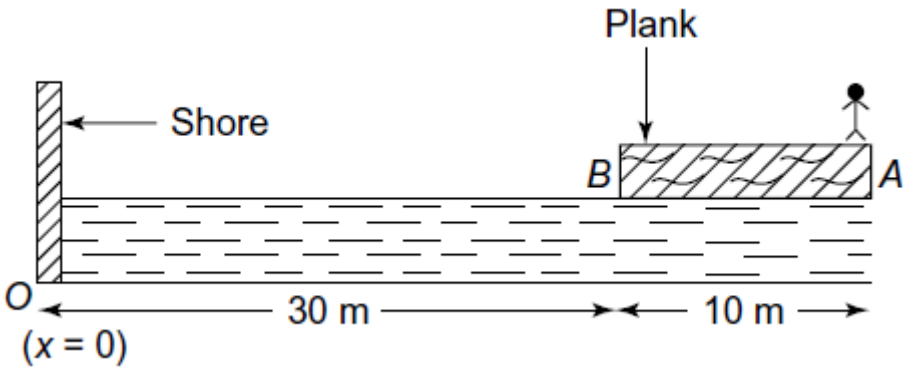
(B) $18\ \text{ms}^{-1}$

(C) $20\ \text{ms}^{-1}$

(D) $17\ \text{ms}^{-1}$

Q.6

A boy of mass $m = 50\ \text{kg}$ stands at the end A of a flat plank AB of wood of mass $M = 100\ \text{kg}$ and length $l = 10\ \text{m}$ floating in the still water in a lake. The end B of the plank is at a distance of $30\ \text{m}$ from the shore of the lake as shown in Fig. 5.10. The boy walks a distance of $6\ \text{m}$ on the plank towards the shore. How far is the boy from the shore now? Neglect viscosity of water.



(A) 36 m

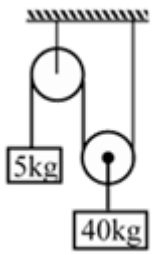
(B) 38 m

(C) 28 m

(D) 27 m

Q.7

The acceleration of centre of mass of the system shown in figure will be



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(A) $10\ \mathrm{m/s^2}$

(B) $-\frac{10}{3}\ \mathrm{m/s^2}$

(C) $\frac{5}{3}\ \mathrm{m/s^2}$

(D) $-5\ \mathrm{m/s^2}$

Q.8

A bullet of mass m moving with velocity v strikes a suspended wooden block of mass M. If the block rises to a height h, the initial velocity of the block after collision will be

(A) $\sqrt{2gh}$

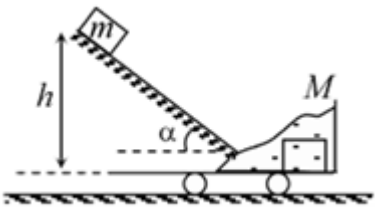
(B) $\frac{M+m}{m}\sqrt{2gh}$

(C) $\frac{m}{M+m}2gh$

(D) $\frac{M+m}{M}\sqrt{2gh}$

Q.9

A block of mass m slides without friction down a fixed inclined board of inclination α with the horizontal. After leaving the inclined, the block falls on a cart of mass M. Initial height of the block above the level of the cart is h as shown. The velocity of cart when block drops on it will be



(A) $\frac{m\sqrt{2gh}}{M+m}$

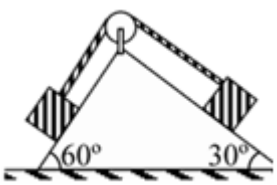
(B) $\frac{m\sqrt{2gh}\sin\alpha}{M+m}$

(C) $\frac{m\sqrt{2gh}\cos\alpha}{M+m}$

(D) $\frac{m\sqrt{2gh}\cos\alpha}{M}$

Q.10

Two blocks of equal mass are tied with a light string, which passes over a massless pulley as shown in figure. The magnitude of acceleration of centre of mass of both blocks is: (wedge is fixed and smooth)



(A) $\left(\frac{\sqrt{3}-1}{4\sqrt{2}}g\right)$

(B) $(\sqrt{3}-1)g$

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(C) $\frac{g}{2}$

(D) $\left(\frac{\sqrt{3}-1}{\sqrt{2}}\right) g$

Q.11

A body A of mass M while falling vertically downwards under gravity breaks into two parts; a body B of mass $\frac{1}{3}\mathrm{M}$ and a body C of mass $\frac{2}{3}$ M. The centre of mass of bodies B and C taken together shifts compared to that of body A

(A) does not shift

(B) depends on height of breaking

(C) towards body B

(D) towards body C

Q.12

Two bodies of masses $2\ \mathrm{kg}$ and $4\ \mathrm{kg}$ are moving with velocities $2\ \mathrm{m/s}$ and $10\ \mathrm{m/s}$ respectively along same direction. Then the velocity of their centre of mass will be

(A) $8.1\ \mathrm{m/s}$

(B) $7.3\ \mathrm{m/s}$

(C) $6.4\ \mathrm{m/s}$

(D) $5.3\ \mathrm{m/s}$

Q.13

Two objects P and Q initially at rest move towards each other under mutual force of attraction. At the instant when the velocity of P is v and that of Q is $2\mathrm{v}$, the velocity of centre of mass of the system is

(A) v

(B) $3\mathrm{v}$

(C) $2\mathrm{v}$

(D) zero

Q.14

Two particles which are initially at rest, move towards each other under the action of their internal attraction. If their speeds are v and $2v$ at any instant, then the speed of centre of mass of the system will be:

(A) $2v$

(B) zero

(C) 1.5

(D) v

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Q.15

A projectile of mass m is thrown with velocity v making an angle of 30° with vertical. Neglecting air resistance the magnitude of change in momentum between the starting point and at the maximum height is

(A) $\frac{mv}{2}$

(B) $\frac{\sqrt{3}mv}{2}$

(C) mv

(D) $\frac{\sqrt{7}mv}{2}$

Answers & Solutions

Q.1 Answer:

8 cm

Solution:

Let $u\text{cms}^{-1}$ be the speed of the bullet. Since the mass of the bullet remains unchanged, its speed becomes $v = \frac{3u}{4}\text{cms}^{-1}$ after it penetrates a distance $x = 3.5$ cm. The retardation a due to the resistance of the wooden is given by

$u^2 - v^2 = 2ax$

or $u^2 - \left(\frac{3u}{4}\right)^2 = 2a \times 3.5$

which gives $a = \frac{u^2}{16}\text{cms}^{-2}$. The bullet will come to rest when its velocity $v' = 0$. If x' is the thickness penetrated by the bullet, then

$u^2 - v'^2 = 2ax'$

or $x' = \frac{u^2}{2a}$. But $a = \frac{u^2}{16}\text{cms}^{-2}$. Therefore

$x' = \frac{u^2 \times 16}{2u^2} = 8$ cm

Hence the correct choice is (a).

Q.2 Answer:

$\frac{m^2v^2}{2(M+m)}$

Solution:

Initial momentum (p) = momentum of bullet + momentum of block $= mv + 0 = mv$. From the principle of conservation of momentum, final momentum of bullet + block system of mass $(M+m)$ = Initial momentum p. Now

$\text{KE} = \frac{p^2}{2\times(M\times m)} = \frac{m^2v^2}{2(M+m)}$

Hence the correct choice is (d).

Q.3 Answer:

$5.2\,\hat{\imath}\text{ms}^{-1}$

Solution:

Since the system (boy + plank) is initially at rest, its momentum is zero. Since no external force acts on

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the system, the momentum of the system will remain zero. Let us assume that the boy runs in the positive x-direction.

Velocity of boy relative to the plank is

$\mathrm{v}_{\mathrm{bp}} = \mathrm{v}\hat{\mathrm{i}}$

Now $\mathrm{v}_{\mathrm{bp}} = \mathrm{v}_{\mathrm{b}} - \mathrm{v}_{\mathrm{p}}$

where $\mathrm{v}_{\mathrm{b}} =$ velocity of boy relative to ground

and $\mathrm{v}_{\mathrm{p}} =$ velocity of plank relative to ground

Hence $\mathrm{v}_{\mathrm{b}} = \mathrm{v}_{\mathrm{bp}} + \mathrm{v}_{\mathrm{p}}$

$= v\hat{\mathrm{i}} + v_{\mathrm{p}}\hat{\mathrm{i}}$

$= (v + v_{\mathrm{p}})\,\hat{\mathrm{i}}$

Total momentum of plank + boy $= 0$

$\Rightarrow \quad M\mathrm{v}_{\mathrm{p}} + m\mathrm{v}_{\mathrm{b}} = 0$

$\Rightarrow \quad Mv_{\mathrm{p}}\hat{\mathrm{i}} + mv_{\mathrm{b}}\hat{\mathrm{i}} = 0$

$\Rightarrow Mv_p\hat{\mathrm{i}} + m\,(v + v_p)\,\hat{\mathrm{i}} = 0$

which gives $v_p = -\frac{mv}{M+m}$

∴ Velocity of boy is $\mathrm{v}_b = \left(v - \frac{mv}{M+m}\right)\hat{\mathrm{i}} = \frac{Mv}{M+m}\hat{\mathrm{i}}$

$= \left(\frac{260\times 6}{260+40}\right)\hat{\mathrm{i}}$

$= 5.2\hat{\mathrm{i}}\,\mathrm{ms}^{-1}$

Q.4 Answer:

$\frac{2}{R}\sqrt{\frac{gh}{3}}$

Solution:

Let M be the mass of the cylinder and R be its radius. When it is at the top of the inclined plane of height h, its potential energy is Mgh (Fig. 5.65). As it rolls down the inclined plane, it moves along the plane and also rotates about an axis passing through its axis perpendicular to the plane of the figure, which shows its section in the plane of the paper. It thus acquires both kinetic energy of translation $\left(\frac{1}{2}Mv^2\right)$ and kinetic energy of rotation $\left(\frac{1}{2}I\omega^2\right)$ where v is its linear velocity and ω its angular velocity when it reaches the bottom of the plane. I is its moment of inertia about the axis mentioned above which is given by

$I = \frac{1}{2}MR^2$

From the law of conservation of energy, we have Potential energy = Translational kinetic energy + Rotational kinetic energy

Or $Mgh = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2$

or $Mgh = \frac{1}{2}MR^2\omega^2 + \frac{1}{2}\left(\frac{1}{2}MR^2\right)\omega^2$

$(\because v = R\omega)$

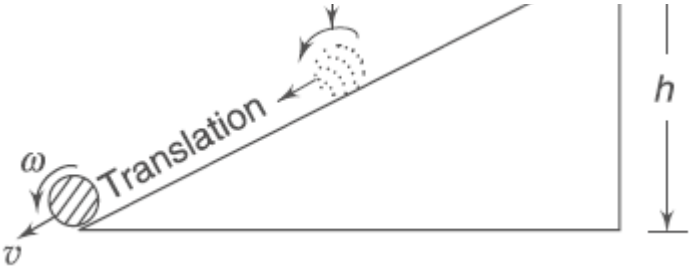
or $= \frac{3}{4}MR^2\omega^2$

$\omega^2 = \frac{4gh}{3R^2}$

or $\omega = \frac{2}{R}\sqrt{\frac{gh}{3}}$**, which is choice (c).**



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Q.5 Answer:

$13\ \mathrm{ms}^{-1}$

Solution:

$V_{\mathrm{CM}}=\frac{m_1v_1+m_2v_2}{m_1+m_2}=\frac{1000\times10+1500\times15}{1000+1500}$
$=13\ \mathrm{ms}^{-1}$

Q.6 Answer:

36 m

Solution:

Initially the centre of mass of the system (plank + boy) is at rest. To walk, the boy exerts a force in the backward direction. The plank in turns exerts a reaction force on the boy in the forward direction. These forces are internal to the system. Since no external force acts, the centre of mass of the system remains at rest even when the boy walks on the plank. Let the shore be at the origin $O(x=0)$. Initially let x be the distance of the centre of mass of the plank from O. Then the distance of the centre of mass of the system (plank + boy) from O will be

$$x_{CM}=\frac{M\times x+m\times(30+10)}{M+m}=\frac{100x+50\times40}{100+50}$$
$$=\frac{100x+2000}{150}$$

Since the boy moves towards the shore and the centre of mass of the system has to remain at rest, the plank will move away from the shore. If x' is the distance moved by the plank, the distance of the centre of mass from the shore when the boy walks $6\ \mathrm{m}$ on the plank is given by

$$x'_{\mathrm{CM}}=\frac{100\,(x+x')+50\,(40-6+x')}{100+50}$$
$$=\frac{100x+150x'+1700}{150}$$

Since $x_{\mathrm{CM}}=x'_{\mathrm{CM}}$,

$$\frac{100x+2000}{150}=\frac{100x+150x'+1700}{150}$$
$$\Rightarrow x'=\frac{300}{150}=2\ \mathrm{m}$$

$\therefore$ **Distance of the boy from the shore** $=40-6+2=36\ \mathrm{m}$

Q.7 Answer:

$-\frac{10}{3}\ \mathrm{m/s^2}$

Solution:

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By constraint relation,

If $a_{40} = a$**, then** $a_5 = 2a$

$40g - 2T = 40a$ **...(i)**

$T - 5g = 5(2a)$ **...(ii)**

From (i) and (ii) $\quad a = 5\ \mathrm{m/s^2}$

$a_{\mathrm{cm}} = \frac{40(-5)+5(10)}{40+5} = -\frac{10}{3}\ \mathrm{m/s^2}$

Q.8 Answer:

$\frac{M+m}{m}\sqrt{2gh}$

Solution:

$\frac{1}{2}(m+M)\left\{\frac{mv}{m+M}\right\}^2 = (m+M)gh$

$\therefore \quad V = \frac{m+M}{m}\sqrt{2gh}$

Q.9 Answer:

$\frac{m\sqrt{2gh}\cos\alpha}{M+m}$

Solution:

Speed of block at the bottom of board $= \sqrt{2gh}$

Applying conservation of linear momentum in horizontal direction.

$m\sqrt{2gh}\cos\alpha = (M+m)v$

$v = \frac{m\sqrt{2gh}\cos\alpha}{M+m}$

Q.10 Answer:

$\left(\frac{\sqrt{3}-1}{4\sqrt{2}}g\right)$

Solution:

Acceleration of system $a = \frac{mg\sin 60^\circ - mg\sin 30^\circ}{2m}$

Here, m **= mass of each block** $\quad$ **or** $a = \left(\frac{\sqrt{3}-1}{4}\right)g$

Now $\vec{a}_{\mathrm{com}} = \frac{m\vec{a}_1 + m\vec{a}_2}{2m}$

Here, $\vec{a}_1$ **and** $\vec{a}_2$ **are** $\left(\frac{\sqrt{3}-1}{4}\right)g$ **at right angles.**

Hence, $|\vec{a}_{\mathrm{com}}| = \frac{\sqrt{2}}{2}a = \left(\frac{\sqrt{3}-1}{4\sqrt{2}}\right)g$

Q.11 Answer:

does not shift

Solution:

Does not shift as no external force acts. The centre of mass of the system continues its original path. It is only the internal forces which comes into play while breaking.

Q.12 Answer:

$7.3\ \mathrm{m/s}$

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Solution:

$\vec{v}_{cm} = \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1 + m_2} = \frac{2\times 2 + 4\times 10}{2+4} = 7.3\ \mathrm{m/s}$

Q.13 Answer:

zero

Solution:

Centre of mass is at rest $\therefore V_{CM} = 0$

Q.14 Answer:

zero

Solution:

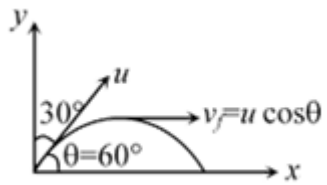
If no external force acts on a system of particles, the centre of mass remains at rest. So, speed of centre of mass is zero.

Q.15 Answer:

$\frac{\sqrt{3}mv}{2}$

Solution:

$\vec{v}_i = u\cos\theta\hat{i} + u\sin\theta\hat{j}$
$\vec{v}_f = u\cos\theta\hat{i}$
$\Delta\vec{p} = m\left(\vec{v}_f - \vec{v}_i\right)$
$\Delta\vec{p} = -mu\sin\theta\hat{j}$
$|\Delta\vec{p}| = mu\sin 60^\circ = \frac{\sqrt{3}mu}{2}$



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